Nisha Menon

Robotics & Control Engineer

RESUME_1683

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Total Experience: 1 year

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 1 year of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: C++, PLC, Python, Gazebo, ROS 2, MATLAB, Simulink, Robot Kinematics, PID Control, Embedded Systems, SLAM, Motion Planning

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Robotics & Control Engineer | Cedarwave Systems | Jul 2025 - Present

- Lowered navigation recovery events by 39% using improved obstacle handling
- Validated 426 robot simulation scenarios with recorded sensor logs
- Improved pick-position repeatability by 25% after calibration updates

PROJECTS

Autonomous Mobile Robot | Gazebo, C++, Embedded Systems, SLAM

- Built ROS 2 nodes for localization, obstacle avoidance and route execution.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Vision-Guided Pick Cell | Python, Simulink, SLAM, Robot Kinematics

- Integrated camera calibration, pick pose estimation and manipulator trajectories.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Western Plains Institute of Technology, 2025
Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

Industrial Robot Safety (2026)